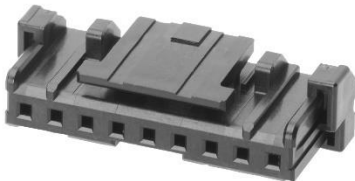


Micro-Lock Plus 2.0

WIRE TO WIRE CONNECTOR SYSTEM (SINGLE ROW TIN-PLATING)

Receptacle Terminal	Receptacle Housing
	
Series: 505572	Series: 505570

Plug Terminal	Plug Housing
	
Series: 213720	Series: 213719

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1.0 SCOPE

This product specification covers the performance requirements for Micro-Lock Plus 2.0 WIRE TO WIRE CONNECTOR SINGLE ROW TIN PLATING series.

2.0 PRODUCTS DESCRIPTION

2.1 DESCRIPTIONS, SERIES NUMBER, AND LINKS

DESCRIPTIONS	PART NUMBER	DRAWING NUMBER
Receptacle terminal	505572 1000	5055721000 PSD 000
Receptacle housing	505570 **0*	5055700001 PSD 000
Plug Terminal	213720 1000	2137200000-SD PSD 000
Plug Housing	213719 2***	2137190000-SD PSD 000

2.2 DIMENSIONS, MATERIALS, PLATINGS

See sales drawings for details on dimensions, materials, platings, and markings.

2.3 ENVIRONMENTAL CONFORMANCE:

- [Go to molex.com](#)
- Enter the part number in the search field.
- At the bottom of the page go to "Environmental" to see compliance status.

2.4 SAFETY AGENCY LISTINGS

UL / cUL File Number: E29179

3.0 REFERENCE DOCUMENTS

3.1 MOLEX DOCUMENTS

Micro-Lock Plus 2.0 w-t-b and w-t-w Connector System Application Specification 5055700001 PS A03
ATS – Application Tooling Specification*

**Application Tooling Specification for terminals is not provided in this document. ATS for terminals can be available from respective terminal part number page in Molex.com*

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4.0 ELECTRICAL PERFORMANCE RATINGS**4.1 VOLTAGE**

250 V AC rms / DC

4.2 RATED CURRENT AND APPLICABLE WIRES

Wire Size	Rated Current (MAX.)	Insulation O.D. φ 0.9~φ 1.5mm
AWG #22	3.4 A	
AWG #24	3.0 A	
AWG #26	2.6 A	

4.3 CURRENT DERATING

AWG	2-circuits	6-circuits	9-circuits
	Current (A)	Current (A)	Current (A)
22	4.7	4.0	3.4
24	3.9	3.1	3.0
26	2.9	2.7	2.6

1. Values are for REFERENCE ONLY.
2. Current deratings are based on not exceeding 30 °C Temperature Rise
3. Temperature Rise is measured in barrel area of crimp terminal.
4. PCB trace design can greatly affect temperature rise results.
5. Data is for all circuits powered.

4.4 TEMPERATURE

Ambient Temperature Range*1*2*3 :- 40 °C ~ + 105 °C
Not freeze to low temperature

NOTE:

- *1. Non-operating connectors after reflow must follow the operating temperature range condition.
- *2. Including terminal temperature rise.
- *3. Applicable wires must also meet the specified temperature range.

4.5 DURABILITY

Plating Type	Number of Cycles
Tin Plated	30 cycles

5.0 QUALIFICATION

Sample selection is in accordance with EIA-364-1000.

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6.0 PERFORMANCE

6.1 ELECTRICAL PERFORMANCE

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT	
6.1.1	Contact Resistance	Mate connectors and measured by dry circuit, 20mV MAX., 10 mA MAX. and subtract wire conductor resistance. Refer to paragraph 8 Contact resistance measuring point. (JIS C5402-2-1)	20 milliohms MAX.	
6.1.2	Insulation Resistance	Mate connectors and apply 500 V DC between adjacent terminals or between terminal and ground. (JIS C5402-3-1 / MIL-STD-202 Method 302)	1000 Megohms MIN.	
6.1.3	Dielectric Strength	Mate connectors, apply 800V AC rms for 1 minute between adjacent terminals or between terminal and ground. (JIS C5402-4-1/MIL-STD-202 Method 301)	No Damage on function	
6.1.4	Contact Resistance on crimped portion	Crimp the applicable wire to the terminal, measured by dry circuit, 20 mV MAX., 10 mA. MAX.	5 milliohms MAX.	
6.1.5	Temperature Rise	Mate connectors and all crimp terminals shall be connected in a direct series. The temperature rise shall be measured when maximum rated current is flowed and thermal equilibrium reaches. (UL498)	Temperature Rise	30 °C MAX.

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6.2 MECHANICAL PERFORMANCE

ITEM	DESCRIPTION	TEST CONDITION		REQUIREMENT
6.2.1	Insertion and Withdrawal Force	Insert and withdraw connectors at the speed rate of 25 ± 3 mm/minute. The data is measured when the housing lock is removed.		Refer to paragraph 7
6.2.2	Crimping Pull out Force	Fix the crimped terminal to the jig, apply axial pull out force on the wire at the speed rate of 25 ± 3 mm/minute. (JIS C5402-16-4)	AWG #22	39.2 N {4.0 kgf} MIN.
			AWG #24	29.4 N {3.0 kgf} MIN.
			AWG #26	19.6 N {2.0 kgf} MIN.
6.2.3	Crimp Terminal Insertion Force	Insert a crimped terminal into the housing		9.8 N {1.0 kgf} MAX.
6.2.4	Crimp Terminal Retention Force	Apply axial pull out force at the speed rate of 25 ± 3 mm/minute on a crimped terminal assembled in the housing.		25.0 N {2.55 kgf} MIN.
6.2.5	Housing Lock Strength [Positive Lock]	Mate connectors and apply axial pull out force at the speed rate of 25 ± 3 mm/minute.	2-3 circuit	25.0 N {2.55 kgf} MIN.
			4-5 circuit	40.0 N {4.08 kgf} MIN.
			6-9 circuit	80.0 N {8.16 kgf} MIN.

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6.2 MECHANICAL PERFORMANCE CONTINUED

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT	
6.2.6	Repeated Insertion / Withdrawal	Insert and withdraw connectors up to 30 cycles repeatedly at speed rate of 10 cycles per minute maximum.	Contact Resistance	40 milliohms MAX.
6.2.7	Vibration	Mate connectors and conduct test subject to the following vibration conditions, for a period of 2 hours in each of 3 mutually perpendicular axes, passing DC 1mA during the test. Cables should be fixed during the test. Amplitude : 1.5mm P-P Frequency : 10~55~10 Hz in 1 minute. Duration : 2 hours in each X.Y.Z.axes. (JIS C 60068-2-6/ MIL-STD-202 Method 201)	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX.
			Discontinuity	1.0 microsecond MAX.
6.2.8	Mechanical Shock	Mate connectors and conduct test subject to the following shock conditions. 3 shocks shall be applied 6 directions along 3 mutually perpendicular axes [$\pm x$, $\pm y$, $\pm z$, each], passing DC 1 mA current during the test. [Total of 18 shocks] Test pulse : Half Sine Peak value : 490 m/s ² {50 G} Duration : 11 ms (JIS C60068-2-27/ MIL-STD-202 Method 213)	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX.
			Discontinuity	1.0 microsecond MAX.

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6.3 ENVIRONMENTAL PERFORMANCE

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT	
6.3.1	Temperature Cycling	Mate connectors and conduct test subject to the following conditions for 5 cycles. Upon completion of the exposure period, the test specimens shall be left at ambient room temperature for 1 to 2 hours. After that, the measurements shall be performed. 5 cycles of : a) $-40\pm3\text{ }^{\circ}\text{C}$ 30 minutes b) $+105\pm2\text{ }^{\circ}\text{C}$ 30 minutes Shifting time: within 5 minutes (JIS C60068-2-14)	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX
6.3.2	Heat Resistance	Mate connectors and expose to $105\pm2\text{ }^{\circ}\text{C}$ for 96 hours. Upon completion of the exposure period, the test specimens shall be left at ambient room temperature for 1 to 2 hours. After that, the measurements shall be performed. (JIS C60068-2-2/ MIL-STD-202 Method 108)	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX
6.3.3	Cold Resistance	Mate connectors and expose to $-40\pm3\text{ }^{\circ}\text{C}$ for 96 hours. Upon completion of the exposure period, the test specimens shall be left at ambient room temperature for 1 to 2 hours. After that, the measurements shall be performed. (JIS C60068-2-1)	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX
6.3.4	Humidity	Mate connectors and expose to $60\pm2\text{ }^{\circ}\text{C}$, relative humidity 90 to 95 % for 96 hours. Upon completion of the exposure period, the test specimens shall be left at ambient room temperature for 1 to 2 hours. After that, the measurements shall be performed. (JIS C60068-2-78/ MIL-STD-202 Method 103)	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX
			Insulation Resistance	100 Megohms MIN.
			Dielectric Strength	Must meet 6.1.3

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6.3 ENVIRONMENTAL PERFORMANCE CONTINUED

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT	
6.3.5	Salt Spray	Mate connectors and expose to the following salt mist conditions. Upon completion of the exposure period, salt deposits shall be removed by a gentle water wash or dip in running water. After that, the measurements shall be performed. NaCl solution Concentration: 5±1 % Spray time: 48±4 hours Ambient temperature: 35±2 °C (JIS 60068-2-11/ MIL-STD-202 Method 101)	Appearance	No Damage
			Contact Resistance	40 milliohms MAX
6.3.6	SO ₂ Gas	Mate connectors and expose to the conditions of 50±5 ppm SO ₂ gas at ambient temperature 40±2 °C for 24 hours.	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX
6.3.7	NH ₃ Gas	Mate connectors and expose to the conditions of 28 % concentration NH ₃ solution for 40 minutes. [The ratio is 25 ml per 1L]	Appearance	No Damage on function
			Contact Resistance	40 milliohms MAX

() : Reference Standard
{ } : Reference Unit

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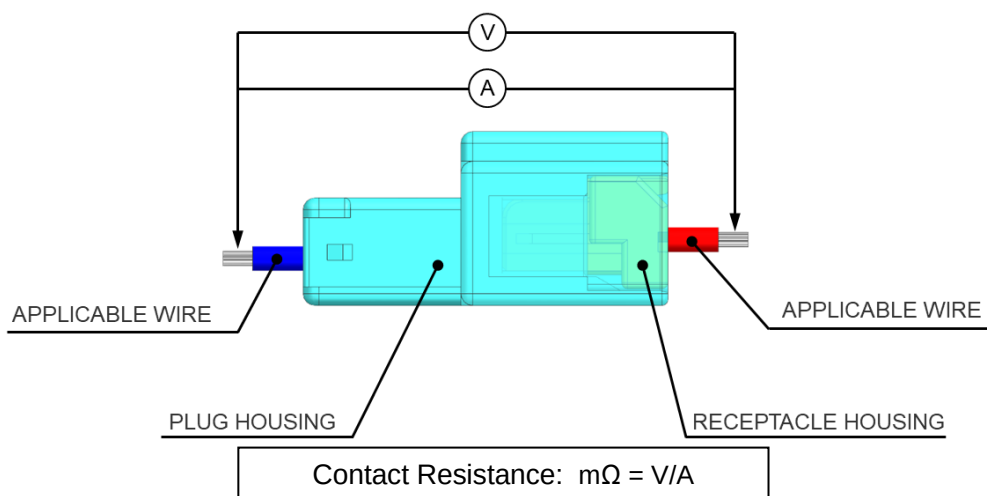
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7.0 INSERTION / WITHDRAWAL FORCE

No. of CKT	UNIT	Insertion (MAX.)			Withdrawal (MIN.)		
		1st	6th	30th	1st	6th	30th
2	N {kgf}	6.5 {0.66}	7.5 {0.77}	14.5 {1.48}	0.3 {0.03}	0.3 {0.03}	0.3 {0.03}
3	N {kgf}	9.7 {0.99}	10.7 {1.09}	17.7 {1.81}	0.5 {0.05}	0.5 {0.05}	0.5 {0.05}
4	N {kgf}	12.9 {1.32}	13.9 {1.42}	20.9 {2.13}	0.7 {0.07}	0.7 {0.07}	0.7 {0.07}
5	N {kgf}	16.1 {1.64}	17.1 {1.74}	24.1 {2.46}	0.9 {0.09}	0.9 {0.09}	0.9 {0.09}
6	N {kgf}	19.3 {1.97}	20.3 {2.07}	27.3 {2.79}	1.1 {0.11}	1.1 {0.11}	1.1 {0.11}
7	N {kgf}	22.5 {2.30}	23.5 {2.40}	30.5 {3.11}	1.3 {0.13}	1.3 {0.13}	1.3 {0.13}
8	N {kgf}	25.7 {2.62}	26.7 {2.72}	33.7 {3.44}	1.5 {0.15}	1.5 {0.15}	1.5 {0.15}
9	N {kgf}	28.9 {2.95}	29.9 {3.05}	36.9 {3.77}	1.7 {0.17}	1.7 {0.17}	1.7 {0.17}

※Release the positive lock, and measure.

{ } : Reference Unit

8.0 CONTACT RESISTANCE MEASURING POINT

9.0 PACKAGING

Parts shall be packaging to protect the parts from damage during standard shipping, storage, and handling. Refer Molex.com specific part number webpage to get the exact packaging document for that item.

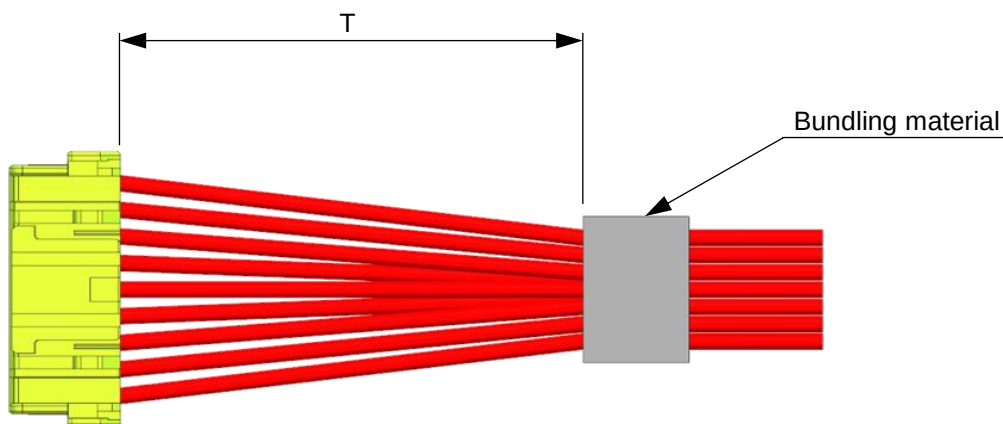
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10.0 CABLE TIE AND / OR TWIST TIE LOCATION

CKT Size	Dim T Min.
2 - 9 ckt	35 mm



The “T” dimension defines a “free” length of wire, or a length of wire that is not subject to significant bias by external factors such as a wire tie, wire twisting, or other means of bending or deforming of the wires that repositions them from their natural relaxed state or location where they enter the housing. Wires are to be dressed in such a manner to allow the terminals to float freely in the pocket. This dimension is general recommendation and may need to be adjusted for different wire gauges and wire type and insulation thickness and insulation material.

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11.0 NOTE

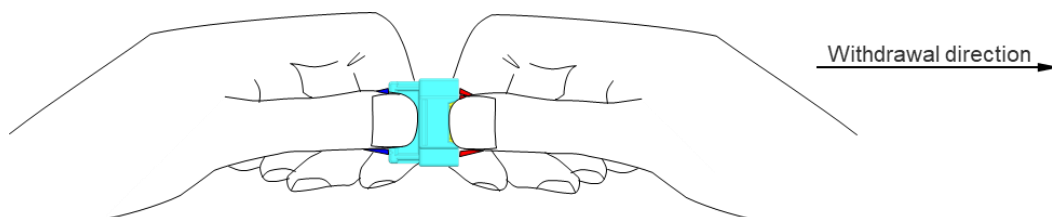
1. When connectors are unmated, positive locks shall certainly be released.
2. There is no influence in the product performance in the case that the black spot or bubble might be appeared on the plastic part of this product and the color shade might be different (including discoloration by aging etc.).
3. The sliding surface marking might be shown on the appearance because the tin plating is used for the tail and nail, however, there is no influence in the product performance.
4. A few scratches may be seen on the surface of the housing and the plating of this product, however, there is no issue in the product performance.
5. Discoloration of the plastic part of this product can be appeared by the exposure to ultraviolet light, however, there is no issue in the product performance.
6. When this product is used at a place where the water exposure could be expected, please provide the appropriate care to avoid the damage from water. There is a possibility of causing insulated malfunction between the circuits by condensation and water leakage.
7. Please do not conduct any washing process on the connectors because it may damage the functionality.
8. Please avoid the situation which the contact area of connector always moves around. For example, the contact area is experiencing a constant movement by the sympathetic vibration of wires and PCB, rotating construction of devices, and action of moveable area. This may cause a defect in the conductivity due to the contact area being worn down. Therefore, please fix wires and PCB on the chassis, and reduces sympathetic vibration.
9. Please do not apply the extra pressure on the connector. For example, do not carry around the substrate which has mated hanging connector on board. There is a case where it causes the damage of connector.
10. After mating the connectors, please do not apply the pressure on the connectors in either the pitch direction, the span direction or rotational direction. It may cause the damage of connector and may crack the soldering.
11. Please ensure to prevent from applying any external forces or shock to the connector when connector or the cable assembly in process or cable assembly is under being packaged, or under transportation. This may cause deformation and damage to the connector and may cause a defect in product performance.
12. When using this product, please ensure to follow within its rated current per circuit. Please ensure not to apply the sum of the rated current separated in several circuits to exceed the maximum allowable rated current.
13. This product is not designed for mating and un-mating of the connector under the condition of an active electrical circuit. It may cause a risk of electric spark and the defect in product performance. Please do not conduct the hot plug and hot unplug.
14. The applicable wire for this connector, in principle, is tin-plated copper stranded wire. Please contact molex in advance when using other kind of wires.
15. Please ensure to keep enough clearance between the connector and chassis of your device in order not to apply pressure on the connector.
16. Please tie the cables at least 35 mm away from the edge of the connector housing and ensure that the forces on all of the wires are applied evenly.

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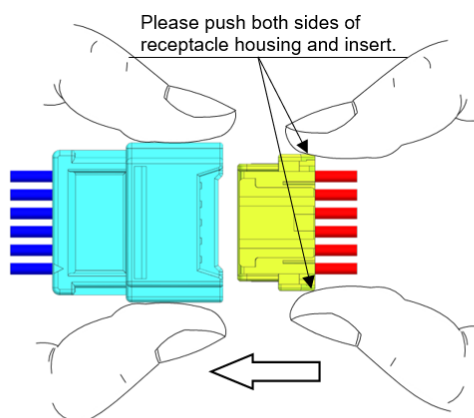

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REV APPR BY	MRAMAKRISHNA	DATE	2023/05/05	PS	PRODUCT SPECIFICATION WORD	000	505570
INITIAL RELEASE				CUSTOMER		DOCUMENT NUMBER	REVISION SHEET
INITIAL DRWN	SOKUMURA01	DATE	2021/03/18	GENERAL	5055700003-PS	A1	12 OF 13
INITIAL APPR	AIDA	DATE	2022/02/18				

17. When extracting a crimp terminal from the housing by using a jig, it may deform the housing lance, and it causes to reduce the terminal retention force enormously after re-inserting the terminal. Therefore, please ensure to use a new housing after repairing the crimp terminals.
18. When positioning cable assembly and cable assembly after mating connectors in the device, it should not have a constant stress or a pulling force applied on it. This phenomenon may damage the contact area, crimping area, or terminal lock area, and it causes the defect in conductivity. Therefore, when designing the wire positioning in the device, please ensure that there is enough wire length not to stress on the connector.
19. When un-mating connector, please ensure to hold wires all together lightly and release the lock completely by attaching flat part of fingers to the lock and pushing the lock bar. Then, please withdraw the receptacle housing slowly and straightly to the mating axial direction. Please avoid withdrawing diagonally and strongly. This may cause the damage of the connector.

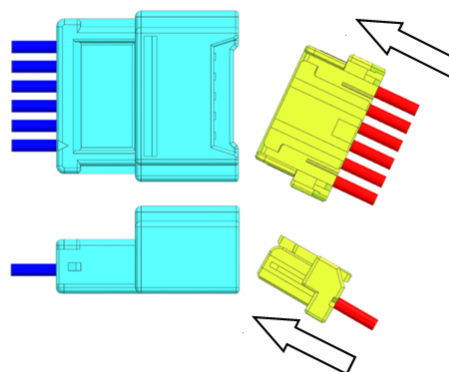


20. Please do not deliberately deform the movable portion (the lock area and lance of housing) and terminals. It would lead to product failure.
21. Please mate the connector in parallel with the mating axis as much as possible. At this time, after positioning and facing each external wall of both receptacle and plug housing, push the connector to mate straight until both plug and receptacle housings strike each other. (This is the complete connector mating position.) In the case of the diagonal mating, position and face lightly each external wall of both receptacle and plug housings less than angle of 10°, and then push to mate. When mating connector in the diagonal angle excessively, it may damage the connector.



Good

Insert straight



No Good

Insert with an angle

22. Because the plug housing material of this product is using polyamide, the water absorption status of the housing material might influence the insertion force / withdrawal force or feeling of insertion. Though its excessive water absorption status may cause to interfere with insertion a bit or to weaken the lock click feeling when mating, it does not damage the product's performance and functions.

[Micro-Lock Plus 2.0 Connector Web Page](#)

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REVISION DESCRIPTION				MICRO-LOCK PLUS 2.0 WIRE TO WIRE CONNECTOR SINGLE ROW TIN-PLATING			
CHANGE NO. 748551				DOC TYPE	DOC TYPE DESCRIPTION	DOC PART	SERIES
REVISED BY	PRAMOD	DATE	2023/05/04	DOC TYPE	PRODUCT SPECIFICATION WORD	000	505570
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